

Assessing the role of mass gathering events on the spread of infectious diseases: RESTART 3.0

Beryl Musundi ¹, Mario Popp ², Stefan Moritz ², Johannes Horn ¹, Aleksandr Bryzgalov ¹, Rafael Mikolajczyk¹, Alexander Kuhlmann ³

¹Martin Luther University Halle-Wittenberg ²University Hospital, Halle (Saale),³University of Lübeck

Introduction

- Mass gathering events (MGEs) can amplify transmission of infectious diseases due to the high density of contacts. Many countries banned MGEs at the onset of the COVID-19 pandemic.
- The RESTART studies [1, 2] demonstrated that appropriate hygiene measures and effective ventilation systems can substantially reduce the risk of airborne virus transmission for indoor MGEs with a seated audience.
- The study aims to assess the differences in contact dynamics between seated and standing audiences at MGEs. These findings can be used to develop more precise models and statements regarding the effectiveness of intervention measures.

Methods

- We organized a live pop concert featuring the german band "Die Prinzen" with 1159 visitors: 583-seated, 576-standing.
- Each participant wore a Contact Tracing Device (CTD) that measured interactions. Contact was recorded when two CTDs were within the distance threshold of 150 cm for at least 30 seconds.
- We analyzed and compared the number of contacts and duration of contact for standing and seated participants across three different concert phases: pre-concert (18:00 to 20:00 CET), concert (20:00 to 22:07 CET), post-concert (22:07 to 22:22 CET) and three time thresholds: >5min, >10min, >15min using R version 4.3.1.
- Data was assessed for normality with median values used for comparison among the scenarios. A negative binomial regression model was used to test for statistical significance between the two groups.

Results

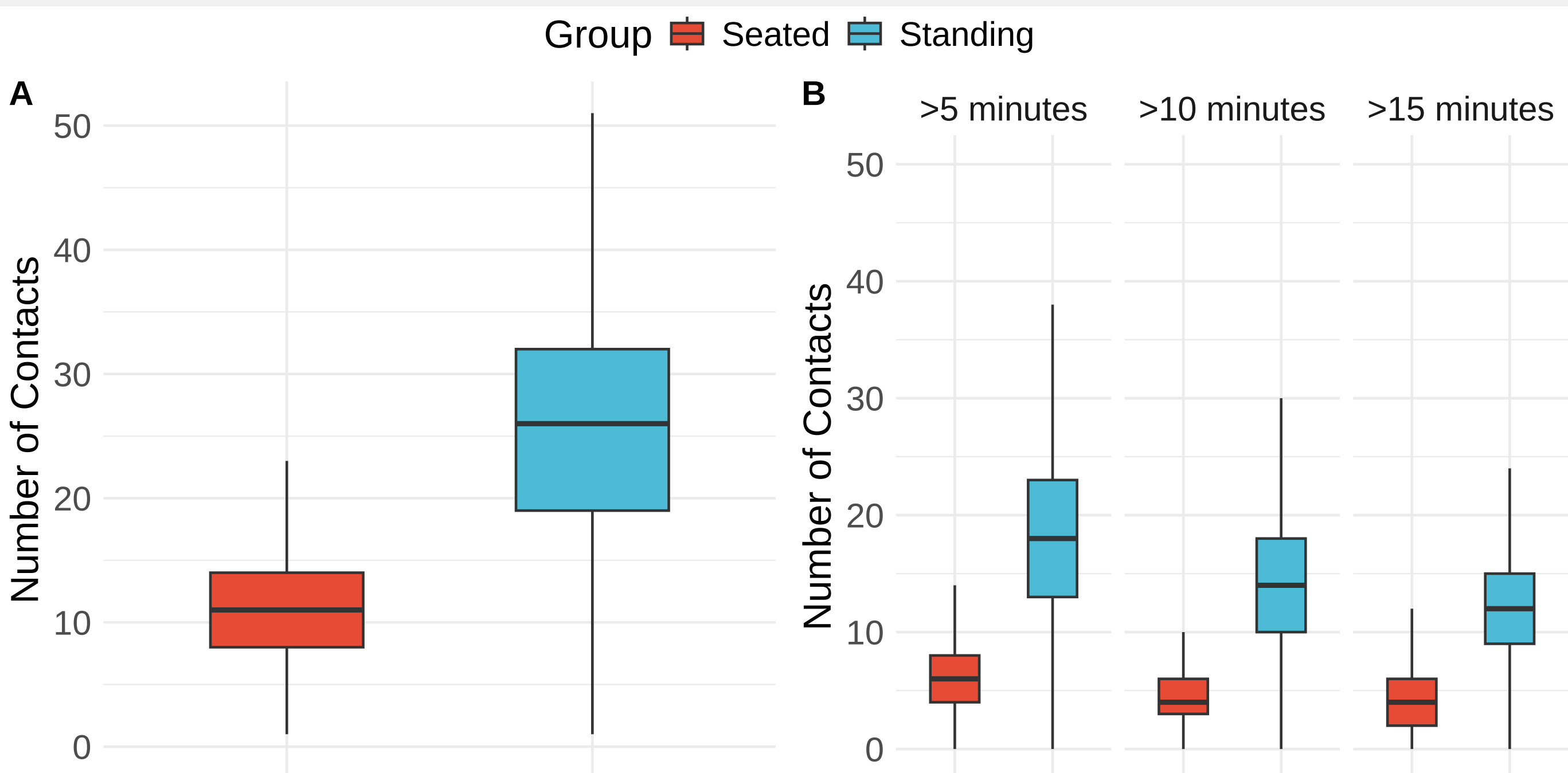


Figure 1. Comparison of the number of contacts in the two groups; A: Overall number of contacts during the whole event; B: Number of contacts in the different time thresholds.

- Standing persons have higher median number of contacts compared to seated persons (Fig. 1A).
- Contacts of seated persons remain stable in the different duration thresholds, while the standing contacts decline with increased contact duration (Fig. 1B).

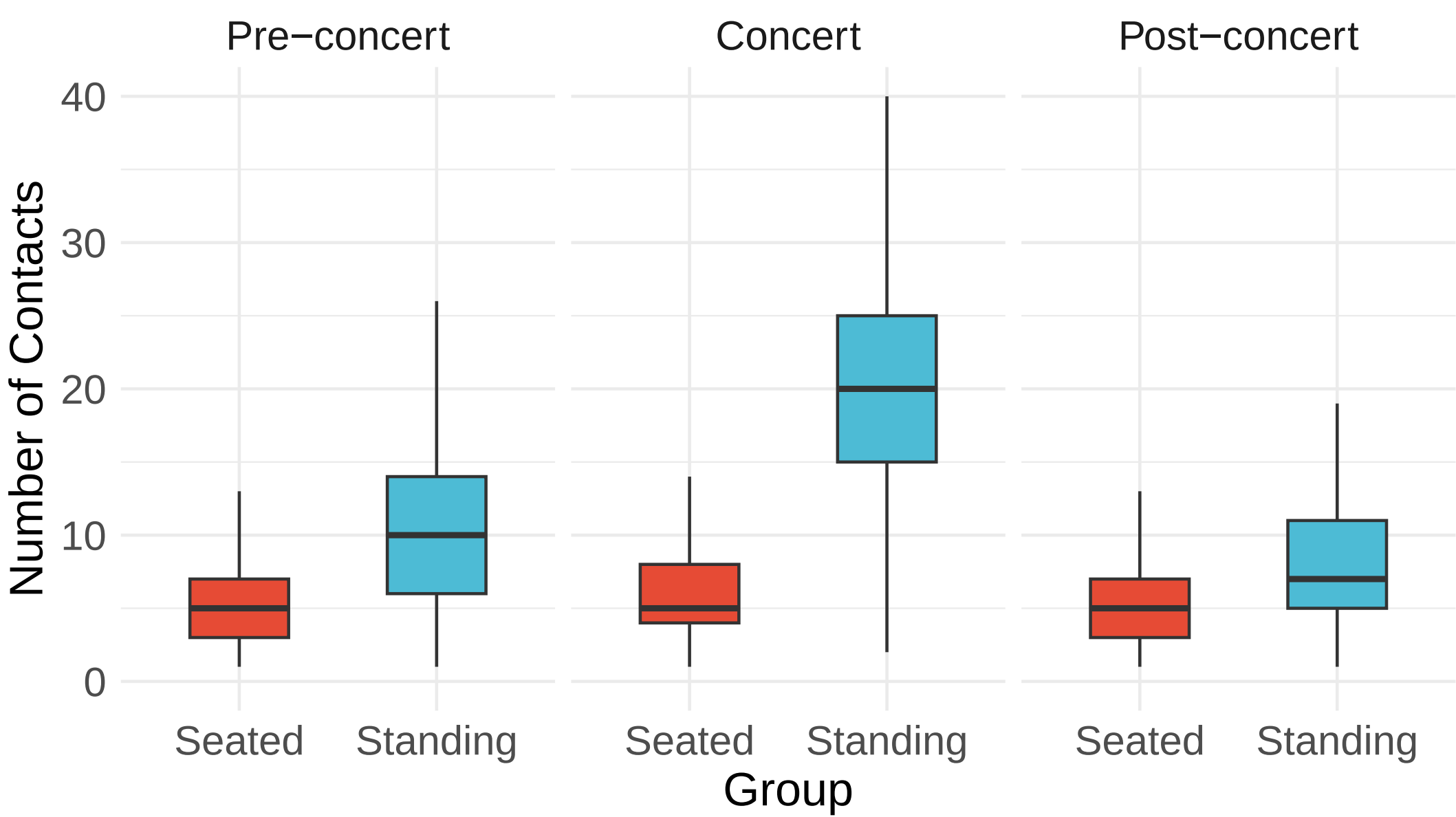


Figure 2. Comparison of the number of contacts in the concert phases.

- Standing persons have the highest median number of contacts in the concert phase while contacts of seated persons remain stable throughout (Fig. 2).

Results

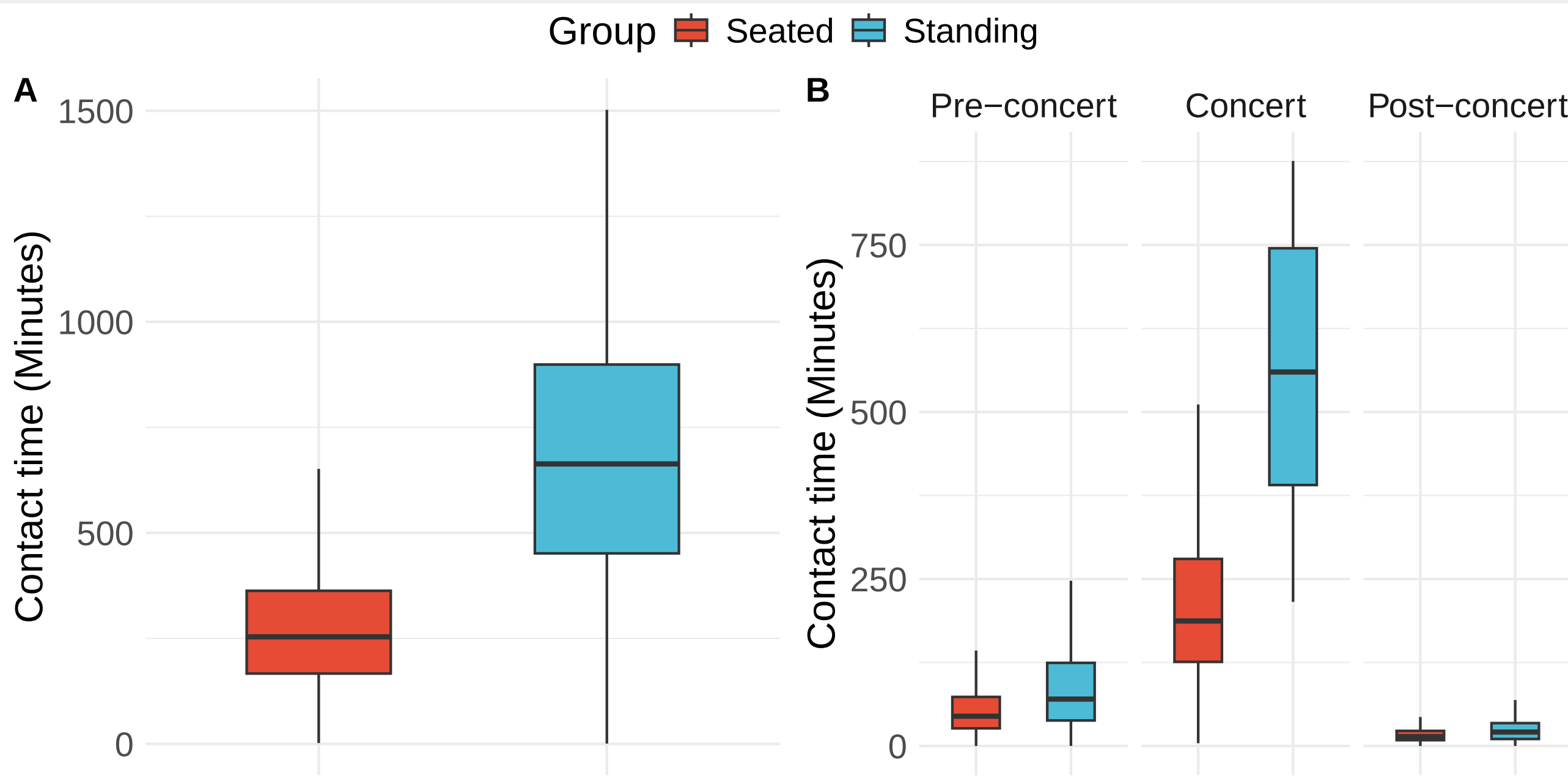


Figure 3. Comparison of cumulative contact time in the two groups: A: Cumulative contact time during the whole event; B: Cumulative contact time in the different concert phases

- Standing persons had overall more median cumulative contact time compared to seated persons (Fig. 3A). The highest contact time was in the concert phase (Fig. 3B).

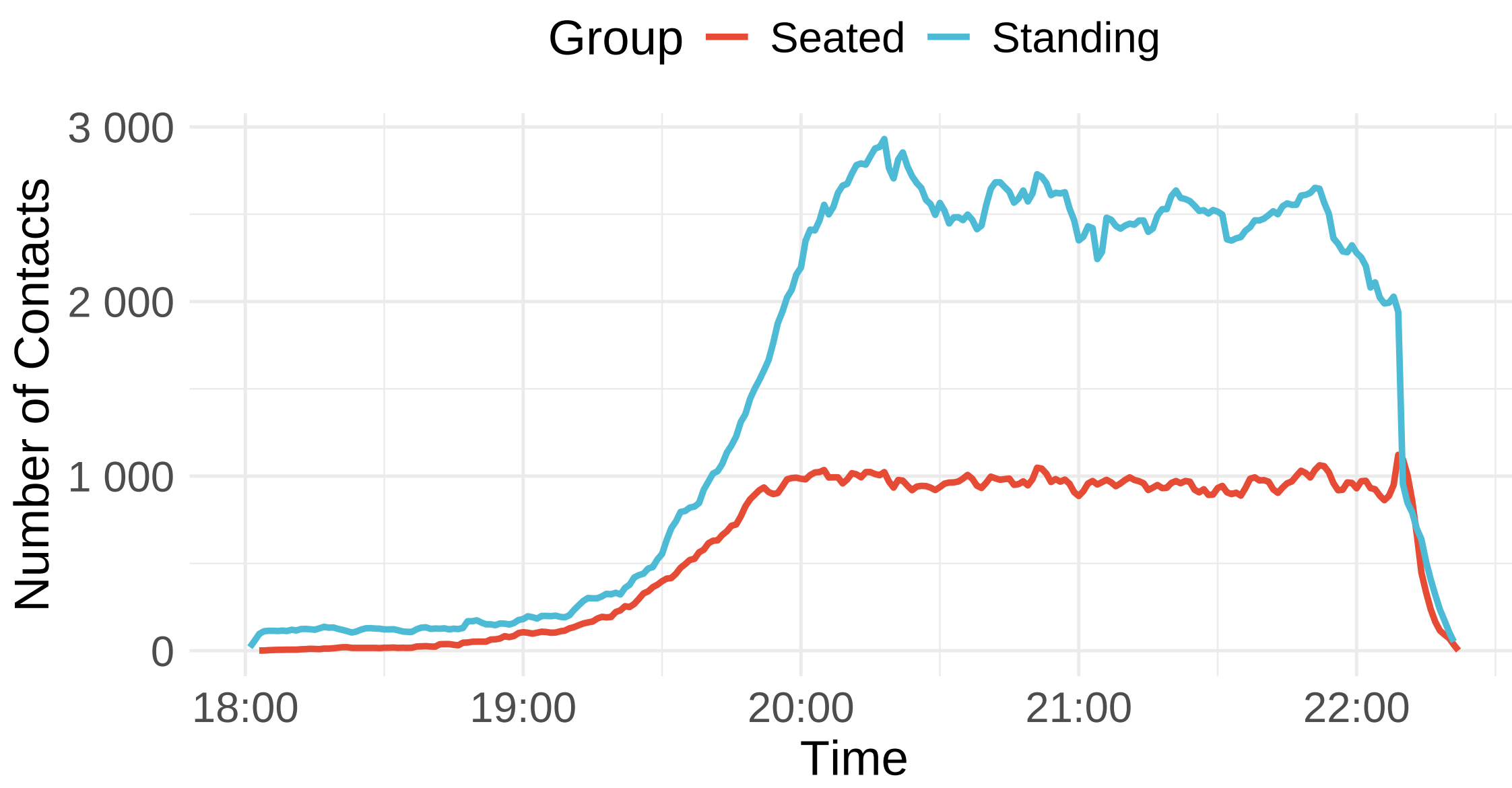


Figure 4. Cumulative contacts at any time during the concert.

- Contacts of standing persons fluctuate during the concert phase while the seated contacts remain relatively stable (Fig. 4).

	Rate ratio [95% CI]			p value		
	Pre-Concert	Concert	Post-Concert	Pre-Concert	Concert	Post-Concert
> 5 min	1.87 [1.72-2.02]	3.19 [2.98-3.40]	1.47 [1.26-1.70]	< 2.2e-16	< 2.2e-16	6.03e-07
> 10 min	1.81 [1.64-1.99]	2.95 [2.75-3.15]	1.78 [1.29-2.47]	< 2.2e-16	< 2.2e-16	4.92e-04
> 15 min	1.80 [1.61-2.02]	2.76 [2.57-2.95]	1.00 [0.00-inf]	< 2.2e-16	< 2.2e-16	1.00

Table 1. Rate ratios comparing the number of contacts for standing vs seated persons.

- There's statistical significance between the two groups with the exception of the number of contacts in the post-concert >15 min threshold (Tab. 1).

Conclusions

- Standing audiences are highly mobile and generate numerous short-term contacts that could be crucial for pathogens with high contagiousity. Hygiene measures would be especially crucial during the concert itself.
- In contrast, seated audiences exhibit limited mobility, with stable low contact rates during the entire concert. However, additional contacts are created at the entry and exit points which would be the crucial time points for hygiene interventions.
- Tailored strategies focusing on the high-contact phases and aerosol management would be essential for MGEs at both seated and standing events.

References

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- S. Moritz, C. Gottschick, J. Horn, et al. The risk of indoor sports and culture events for the transmission of covid-19. *Nature communications*, 12(1):5096, 2021.